

BOARD BRIEFING · 8 OCTOBER 2026

Where AI can speed up travel-expense claims

An AI opportunity assessment for Falcon Bay Services LLC (fictional), a 600-person services firm in Dubai

Sara Hebbadj · portfolio case study · process data: a real public event log used as a stand-in for the client (BPI Challenge 2020, 10,500 claims); interviews, survey and claims are synthetic

Recommendation: pilot a pre-check that warns; fix payments and reminders without AI

12.4%

of claims are rejected at least once. Rejected claims take **11.2 days** (median) instead of 7.3.

40/40

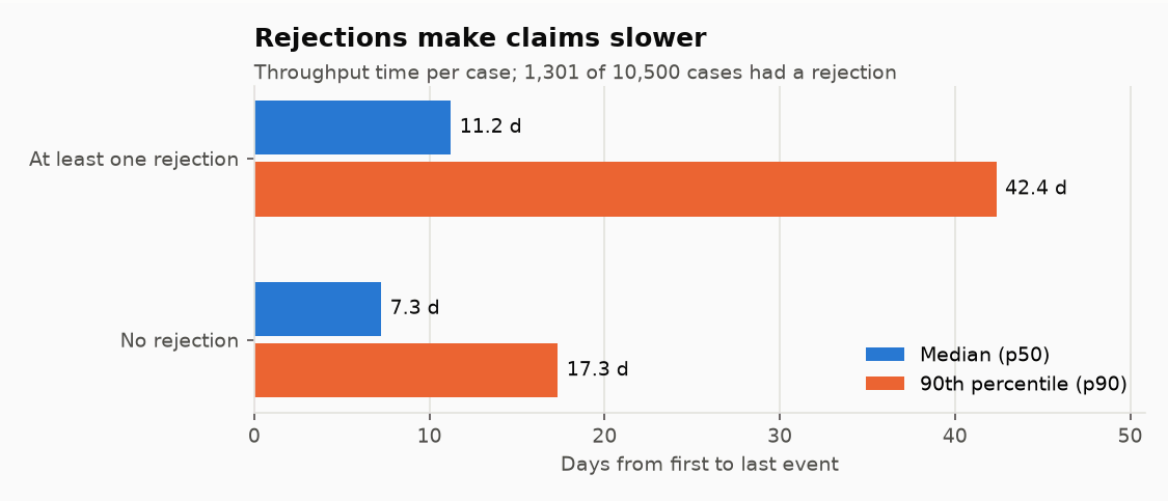
planted policy problems found by rules + gpt-6-luna on 50 test claims, vs 24/40 for rules alone, at US\$0.02 per 100 claims.

56%

of all waiting happens **after** approval, in the payment stage. That is a process fix (pay daily), not an AI project.

- **Pilot UC01** (Receipt and policy pre-check before submission) for 6 weeks in two departments, advisory only. Base-case 3-year net -AED 5k: it only pays back if it prevents at least **42%** of rejections, which the pilot must measure.
- **Switch on the existing approver reminders now** (no AI). Add the AI summary of UC02 (Approver assistant: reminders plus a one-line claim summary) only if a 2-week test shows at least 0.5 minutes saved per approval: base case 3-year net AED 156k, but its rank was not robust (8th–9th for all three model scorers).
- **Decision today:** approve the pilot budget, a UAE-hosted model option, and a daily payment run.

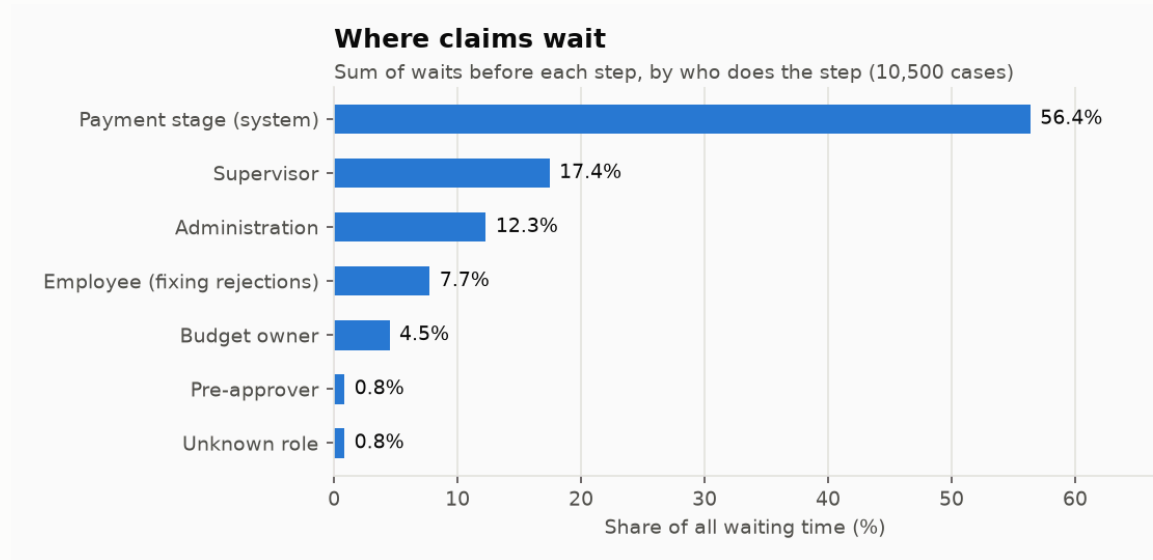
A claim takes a week; one in eight is rejected and takes much longer



Measure (10,500 claims, 2017–18)	Value
Median / p90 throughput	7.3 / 20.1 days
Claims rejected at least once	1,301 (12.4%)
...of which administration rejected first	64%
Claims resubmitted	9.7%
Different process paths	99
Claims in 2018 (annual volume used)	8,260

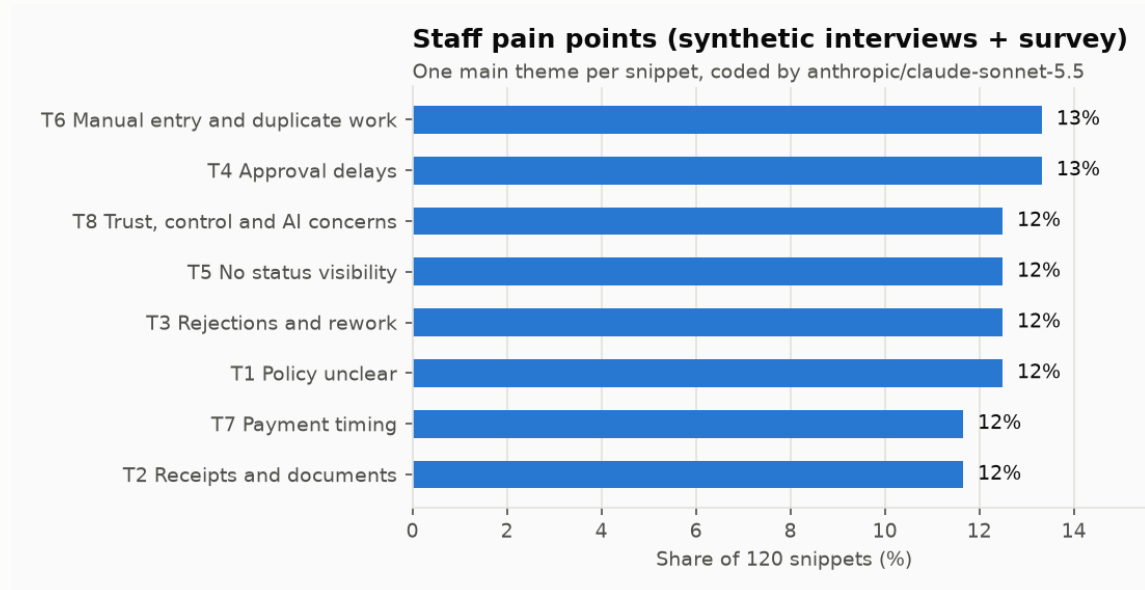
Every number is rebuilt from the raw log by one command; 20 of the calculations are checked against a hand-computed 20-case example (20/20 match).

Most waiting happens after approval, in the payment stage



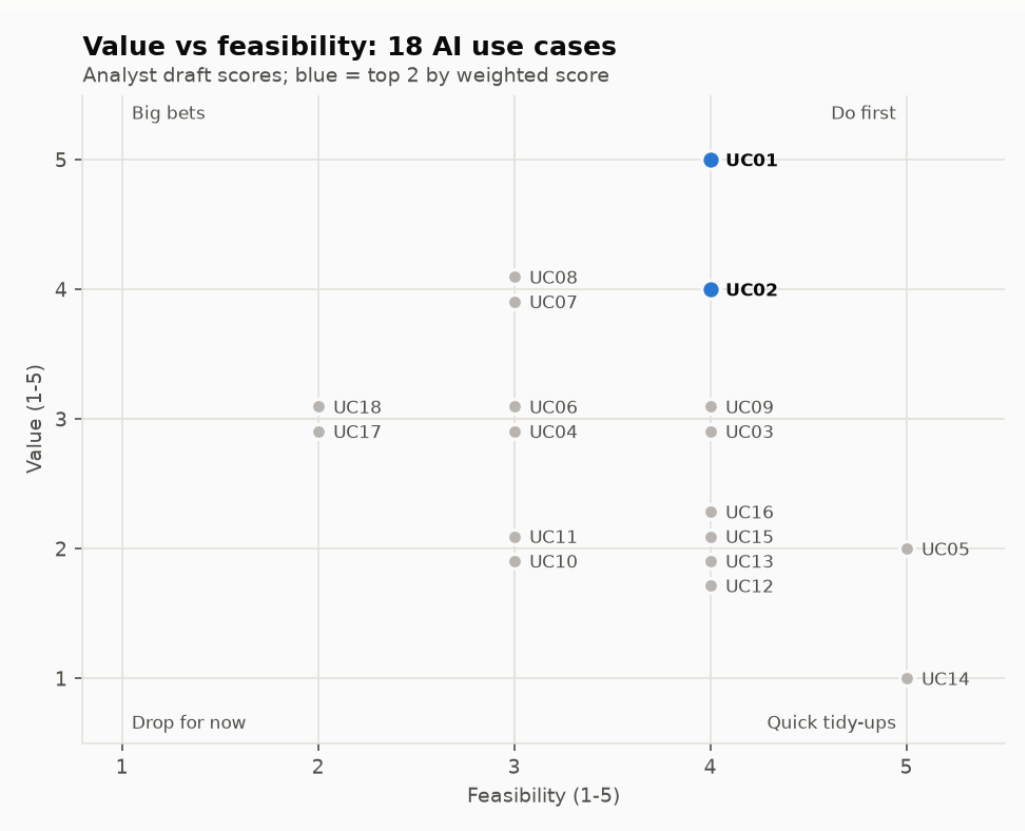
- The payment stage holds **56%** of all waiting time; from payment request to payment takes 3.2 days at the median. Payments run on fixed days: **a daily payment run is the fastest win, and needs no AI.**
- Supervisors are on 97% of claims and hold 17% of waiting; their p90 wait is 5.9 days.
- After a rejection, employees take 1.0 days (median), 5.7 (p90) to respond.

Voice of staff: an LLM codes interviews as consistently as the answer key



- 12 interviews and 40 survey answers (synthetic), split into 120 snippets and coded into 8 themes by an LLM, one snippet per call.
- Agreement with the theme each snippet was written for: gpt-6-luna κ 0.93, gemini-3.8-flash κ 0.96, claude-sonnet-5.5 κ 0.97 (n = 120). This is an upper bound: the same agent wrote the snippets and the labels.
Sara's blind coding is pending.
- Survey (n = 40, 1–5, synthetic): "I can see where my claim is" 2.0; "I would use a pre-check" 3.9.
- What staff asked for: **"warn, never block"** and **"flag the claim, not the person"**.
- The synthetic snippets were written to cover the 8 themes about evenly, so the shares above do not rank pain points; with real interviews they would.

18 AI ideas, scored on value, feasibility, data, risk and speed



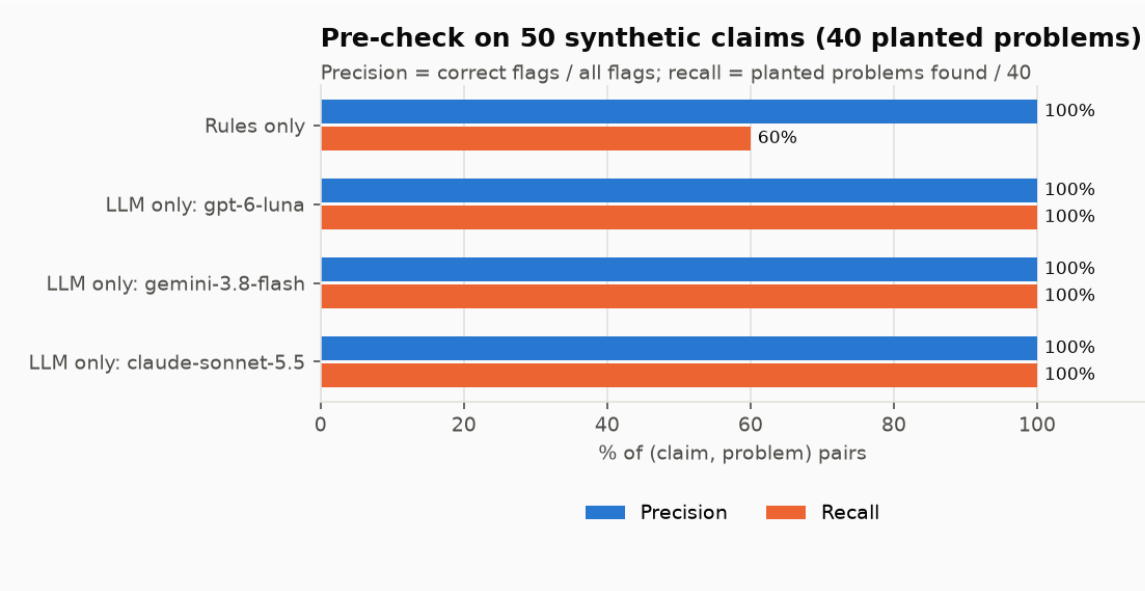
#	Use case (analyst draft)	Score
1	UC01 Receipt and policy pre-check before submission	4.10
2	UC02 Approver assistant: reminders plus a one-line claim summary	4.00
3	UC05 Weekly bottleneck report with an AI-written summary	3.95
4	UC03 Arabic/English travel-policy Q&A assistant	3.70
5	UC14 Draft-claim completion nudges	3.60

Weights: value 30%, feasibility 20%, data readiness 20%, risk 15%, time to value 15%.

Robustness: three LLMs scored every use case blind. Rank agreement with the draft: gpt-6-luna $p = 0.64$, gemini-3.8-flash $p = 0.54$, claude-sonnet-5.5 $p = 0.79$. 0 of 3 kept the same top 2.

All three models put UC06 ("where is my claim?" chat) 1st or 2nd; the draft put it 10th because showing the existing status in the portal needs no AI. UC02 fell to 8th–9th for all three. The rubric has no "is AI needed?" criterion: next version adds one. Lowest everywhere: UC18, a fraud score per employee.

Prototype: rules do the arithmetic, the model reads the receipts



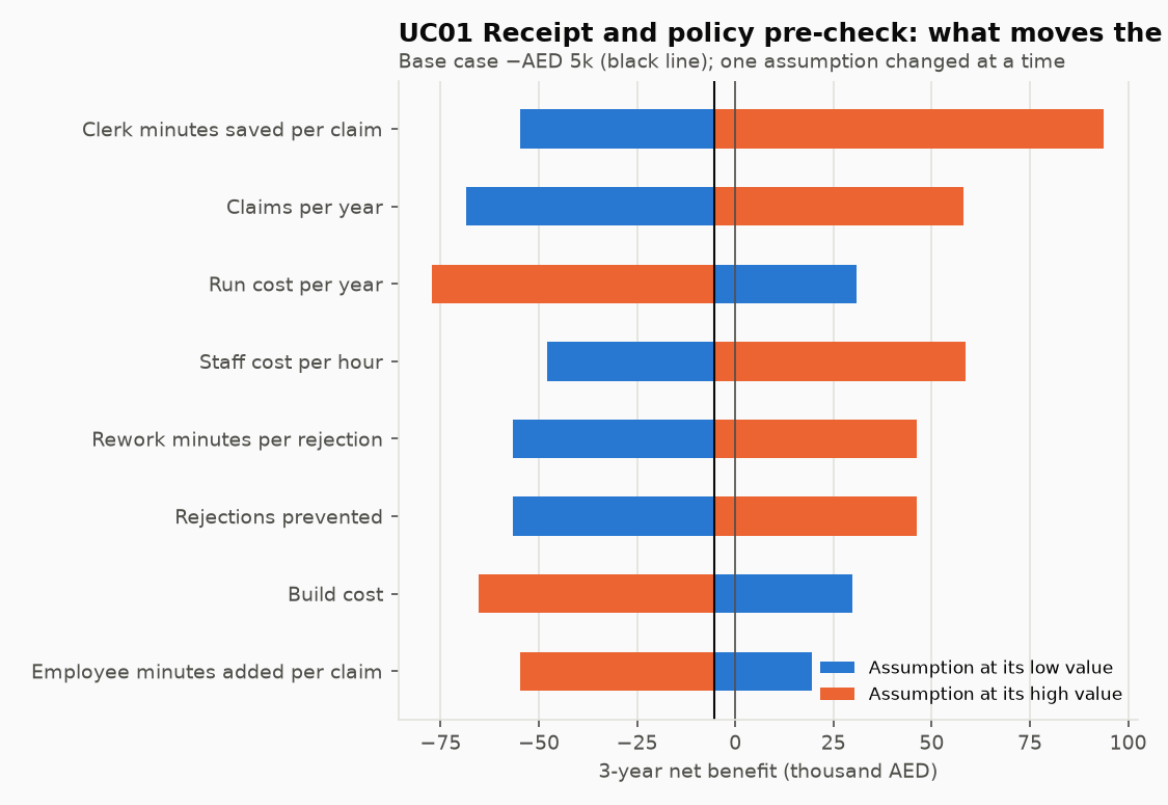
System	Precision	Found	Only in receipt	Per 100 claims
Rules only	100%	24/40	0/16	US\$0.00
LLM only: gpt-6-luna	100%	40/40	16/16	US\$0.02
LLM only: gemini-3.8-flash	100%	40/40	16/16	US\$0.15
LLM only: claude-sonnet-5.5	100%	40/40	16/16	US\$0.53

Rules + each model scored the same as the model alone. All three models hit the ceiling, so this set cannot separate them: the difference is cost (about 34× between the cheapest and the dearest) and speed.

50 synthetic claims, 40 planted problems (24 visible in the form fields, 16 only in the receipt text, some receipts in Arabic). The rules see only the fields by design. Same agent wrote the claims and the rules; one run per model on 8 October 2026.

The business case: UC01 is close to break-even; UC02 depends on one assumption

Use case	Benefit / year	Net / year	3-year net	Payback	Break-even
UC01 pre-check	AED 43k	AED 18k	-AED 5k	no payback in 3 years	8,601 claims/yr or 42% prevented
UC02 approver assistant	AED 80k	AED 65k	AED 156k	7 months	0.5 min saved per approval



- Volumes and rejection rate are measured (8,260 claims in 2018, 13.0% rejected). Model cost per claim is measured in the prototype. Minutes, hourly costs and build costs are **labelled assumptions** with low/high values (docs/roi_model.xlsx).
- For UC01, the clerk minutes saved on every claim move the result most, then volume and run cost; the share of rejections prevented must reach 42%. For UC02, everything rests on approver minutes saved per approval. Faster reimbursement for staff is real but not counted in AED.

Risks are manageable if the AI warns and a person decides

Use case	EU AI Act (benchmark)	Main control
UC01 pre-check	Minimal risk while advisory	Never rejects; flags go to the employee only
UC02 approver assistant	Minimal risk	Summary shows facts, never "approve"
UC08 straight-through approval	Borderline (Annex III 4(b) if ML decides)	Not now
UC18 fraud score per employee	Likely high-risk (Annex III 4(b))	Do not build

- **UAE PDPL** applies (employees' data): keep a human decision (Art. 18), minimise what is sent, and solve the cross-border transfer: IT wants AI inside a UAE cloud tenancy.
- **NIST AI RMF**: owner and change log (Govern), intended use (Map), precision/recall and cost (Measure), weekly flag review and an off switch (Manage).
- General information, not legal advice; see docs/risk_screen.md.

Pilot: 6 weeks, two departments, with kill criteria agreed up front

Week	What happens
0	Baseline rejection rate, data terms, UAE hosting choice, briefing for clerks
1	Shadow mode: the pre-check runs, staff do not see it; clerks label every flag
2-5	Live for the pilot group (warnings only); comparison group as today
3	Checkpoint against the kill criteria
6	Read-out and go / no-go

Stop the pilot if (any one):

- flag precision on clerk-reviewed real claims is below 80% (week 1 or 3);
- it misses more of the clerks' problems than the rules alone;
- receipts leave the approved hosting, or any personal-data incident;
- staff spend more than 2 extra minutes per claim on warnings;
- by week 6 rejections are not trending towards the break-even reduction (42%).

Owner: finance operations lead · **Analyst:** Sara · **Reviewers:** internal audit, IT.